

any shape and size based on system requirements. For this prototype,



Fig. 8: Raspberry Pi and camera module mounted inside the 3D-printed speaker enclosure

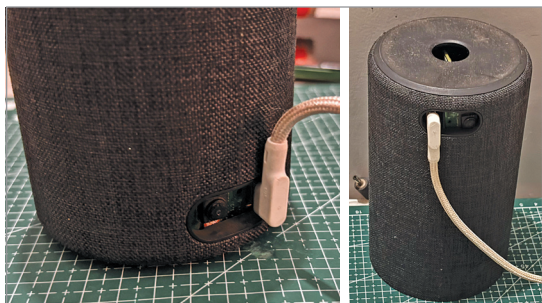


Fig. 9: Camera mounted in the enclosure and USB power connected

a cylindrical design inspired by the second-generation Alexa smart speaker was adopted, with perforated side openings to enable clear audio output. Cut-outs are also provided for the USB power connector and camera module.

After completing the design, 3D-print the enclosure and assemble the hardware inside it. For a professional finish, the perforated section can be covered with speaker mesh fabric or acoustic cloth. Fig. 7 shows the enclosure designed for the generative AI smart speaker.

After 3D-printing the enclosure and attaching the speaker mesh fabric, install the Raspberry Pi inside the enclosure and secure it using the mounting holes. Ensure that the USB-C power connector is properly aligned with the corresponding cut-out in the enclosure. Next, mount the camera module in its designated slot so that the lens is clearly visible through the camera opening. Finally,

place and secure the speakers inside the enclosure.

Once the Raspberry Pi, voice bonnet, camera module, and speakers have been assembled inside the enclosure, the generative AI smart speaker is ready for operation. Fig. 8 shows the Raspberry Pi and camera module mounted inside the 3D-printed speaker enclosure, while Fig. 9 shows the camera mounted in the enclosure and USB power connected to the generative AI smart speaker.

Testing

As explained earlier, the complete software for this system is contained in the Gen_AI_BOX.py program. For testing, open a terminal on the Raspberry Pi, navigate to the system directory, and execute the Gen_AI_BOX.py script.

Once the application launches successfully, it automatically initialises all required services, including audio input/output, camera access, speech processing, and communication with the configured AI models. After the initialisation process is complete, the generative AI smart speaker is ready for testing.

Using the device as a Bluetooth speaker

To use the system as an Alexa-like Bluetooth speaker, first enable Bluetooth on the Raspberry Pi. Then, enable Bluetooth on a smartphone and pair it with the Raspberry Pi. After a successful connection, open the Raspberry Pi audio settings and select voice bonnet or analogue audio output as the audio output device. Set the connected Bluetooth device as the audio input source.

The system will then function as a standard Bluetooth speaker, allowing music, videos, podcasts, and other audio content to be streamed from the paired smartphone, similar to a commercial smart speaker. Fig. 10 shows the Raspberry Pi audio settings used during testing.

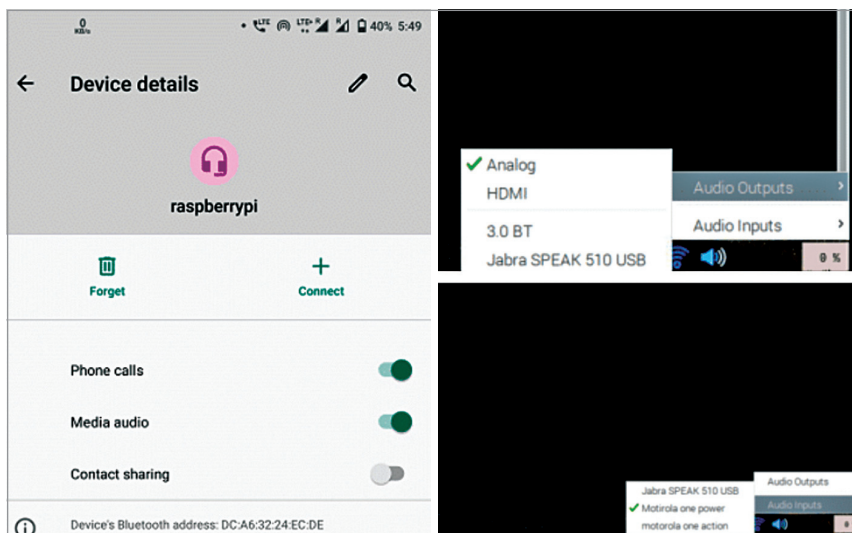


Fig. 10: Raspberry Pi audio settings